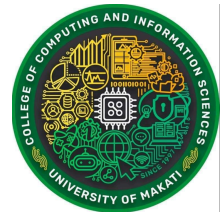




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Vulnerability Assessment Report

A Final Term Requirement for The Course INFORMATION AND ASSURANCE 1

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I. INTRODUCTION AND SCOPE

The assessment involves systematic evaluation of the security posture of participating organizations by identifying existing vulnerabilities, measuring risk exposure, and benchmarking their practices against industry peers. The assessment involves operations focusing on IT-enabled environments such as esports cafes and corporate offices. The team ensures a systematic baseline for comparing vulnerabilities, operational security, and incident management protocols across different IT environments.

The conducted assessment allows the team and focal organization to strengthen technical controls, enhance policy frameworks, and support the development of a long term security strategy.

II. ORGANIZATIONAL PROFILE

HG ESports Cafe – is a newly established business engaged in providing esports and computer rental services. It caters primarily to customers seeking access to online gaming, internet browsing, and other computer-related activities. As a developing organization, it operates in a dynamic environment where consistent service delivery and reliable connectivity are essential to meet user demands.

III. ASSESSMENT METHODOLOGY

To achieve a reliable assessment of the security posture of the four selected organizations, the group implemented a three-phase manual evaluation approach. The assessment centered on examining how each organization manages its operations, safeguards its physical resources, and applies technical security measures. The process was carried out through the following phases:

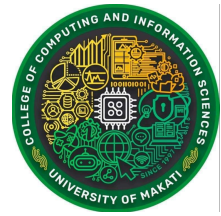
Survey Questionnaire: A standardized Information Assurance and Security questionnaire was administered to the representatives of each organization. Using a



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4-point Likert scale, the instrument collected relevant data regarding organizational practices, security policies, and the level of implementation of security controls.

Management Interview: A follow-up interview was conducted with the representative of the main organization to support and confirm the data gathered from the questionnaire. This phase also provided additional understanding of the organization's operational activities and security-related practices.

Physical & Technical Observation: An onsite observation was performed within the primary organization to examine its working environment. This included reviewing the general setup of its systems and evaluating how physical and operational security measures are applied in practice.

IV. DATA COLLECTION PROCEDURES

Data were collected through a structured survey questionnaire and a supplementary interview with the organization's supervisor/manager. The questionnaire used a 4-point Likert scale and covered domains including IT strategy, security policies, network management, user controls, incident management, and risk management. The interview was conducted to further clarify responses and provide additional insights into the organization's operational practices and security environment.

V. KEY FINDINGS

The assessment of HG Esports Cafe reveals that while certain basic security practices are in place, several critical controls remain informal or partially implemented. Based on the interview and survey responses, the organization operates with updated Windows operating systems and regularly updated software, including weekly system updates. User access is also moderately controlled, as employees have role-based privileges and customers are restricted from administrative functions. However, the absence of a firewall and undisclosed antivirus solution indicates a significant gap in foundational network protection, which may expose the system to external threats.



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In terms of network and user control, HG Esports Cafe demonstrates mixed implementation. The network is partially segmented, as customer devices are kept separate from the server environment, and a separate internet connection is provided for Wi-Fi users when needed. However, firewall protection is not implemented, and users are still allowed to download files under limited supervision. While a reboot-to-restore system is used to clear user data after sessions, the allowance of USB access without detailed safeguards presents a continued risk of malware introduction and system compromise.

From an operational and physical security perspective, the organization shows both strengths and limitations. The server is physically separated from client units but remains exposed within the premises, and no UPS system is in place to mitigate power interruptions. Previous incidents, such as hardware-related black screen issues, highlight vulnerabilities in system stability. Although basic controls such as router password changes and locked physical assets are implemented, the lack of documented security policies, incident protocols, and advanced protective mechanisms such as IDS/IPS or MFA suggests that the organization's overall security posture remains at a developing stage.

VI. ORGANIZATIONAL ASSETS

Table 1. Organizational Profile for HG ESports Cafe

Assets	Hardware / Description	Quantity
Server Infrastructure	LAN Server/On-premise Servers (Locked/Restricted access)	Available



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Router	TP-Link ER605 Multi-WAN Router (supports 3 ISPs, load balancing, failover)	1
Switch	TP-Link TL-SG1024D 24-Port Gigabit Switch (main distribution for PCs)	4
Network Topology	Star Topology (all PCs connected to central switch/router)	Available
Internet / ISP	PLDT-2, Converge, and Globe.	4
Hardware and Networking Components	Uninterruptible Power Supply (UPS) units	Not Available
	Desktop PCs (Elite) (Ryzen 5 5500, 16GB RAM, GTX 1060, 27" Curved 180Hz FHD)	18
	Desktop PCs (VIP) (Ryzen 5 5600, 16GB RAM, GTX 1660 Super, 27" Curved 170Hz 2K)	25



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	Desktop PCs (S-VIP) (Ryzen 5 5600, 16GB RAM, RTX 5060 OC, 27" 240Hz FHD)	27
	Monitors (Elite) (27" Curved 180Hz Full HD monitors)	18
	Monitors (VIP) (27" Curved 170Hz 2K monitors)	25
	Monitors (S-VIP) (27" 240Hz Full HD monitors)	27
	Redragon K686 HE Rapid Trigger Magnetic RGB Gaming Keyboard	70
	Headset (7.1 Badwolf Headset)	70
	Mouse (Rapoo Gaming Mouse (Elite & VIP), Badwolf Gaming Mouse (S-VIP))	70
	Operating System (Windows 11 Home installed on all units)	70
	Alternative Telecommunication Lines	4



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	Fire-prevention (Heat/Smoke detectors/Fire Alarms)	Available
	Alternative Water/Power/Cooling resources	Available
Softwares	Windows 11 home	70
Personnel	Employees	4
Security cameras/CCTV	Not disclosed.	3
Air Conditioner	Carrier 53FLG004-ASC04H - Air Conditioner -2 CARRIER FP-53CXV030308-1 3.0hp Wall Mounted Split Type Aircon -4	6
Data Assets		



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This table presents the asset profile of HG E-Sports iCafe, covering its IT infrastructure, hardware, networking components, software, and personnel. The café uses an on-premise server with restricted access, supported by a TP-Link ER605 multi-WAN router and four (4) TP-Link TL-SG1024D switches arranged in a star topology. This setup enables centralized management and load balancing across four ISPs (PLDT, Converge, and Globe) to maintain stable and reliable connectivity.

The hardware includes a tiered system of Elite, VIP, and S-VIP desktop units, along with standardized peripherals to ensure consistent performance and user experience. Environmental support systems such as air conditioning, fire-prevention mechanisms, and alternative power or cooling resources are also in place to sustain continuous operations and protect physical assets.

All units operate on Windows 11 Home and are managed by a small team of employees, providing an organized and efficient system for both technical and operational functions.

VII. RISK ASSESSMENT AND VULNERABILITY CLASSIFICATION

A. Technical & Network Vulnerabilities

The table below presents security gaps identified within HG E-Sports iCafe's network systems and endpoint devices. These weaknesses could leave the infrastructure vulnerable to threats such as unauthorized system access, malware infiltration, and interruptions to normal service operations.

Table 2. Risk Assessment and Vulnerability Classification for Technical & Network

FINDINGS		POTENTIAL IMPACT	VULNERABILITY CLASSIFICATION	RISK CATEGORY
Black Screen Hardware Malfunction	The company experienced hardware issues.	The most damaging issue has been black screening caused by hardware malfunctions,	High	High



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		leading to downtime and repair costs.		
Multiple ISP Dependency (4 ISPs)	The shop uses two PLDT lines, one Converge, and one Globe connection.	While redundancy improves uptime, misconfiguration or lack of proper load balancing could still cause instability or downtime.	Low	Low
USB Device Access	Users are allowed to plug in their peripherals(usb devices) freely	Without strict controls, USB-borne malware could spread across machines.	High	High
Undisclosed Malicious Downloads	Users are allowed to download anything on the pc.	Customers could bypass the “ask staff first” rule by simply not informing supervisors. This opens the possibility of installing hidden monitoring tools or malware that can operate during a session.	High	High
Recent OS and Software Updates	Systems are updated.	Systems were updated recently (last week).	Low	Low
No Redundancy in Network Infrastructure	The Internet Cafe uses a single router setup.	A single router creates a critical single point of failure. If it fails, all 70 endpoints will lose network connectivity and internet access, resulting in a complete disruption of all operations.	High	High



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B. Physical & Environmental Vulnerabilities

Safeguarding physical assets and maintaining proper environmental controls are essential for ensuring the reliability of the systems that support the business. The following table outlines vulnerabilities in how HG E-Sports iCafe's equipment is secured and maintained, highlighting risks that could result in hardware damage, operational downtime, or theft.

Table 3. Risk Assessment and Vulnerability Classification Physical/Environmental

FINDINGS		POTENTIAL IMPACT	VULNERABILITY CLASSIFICATION	RISK CATEGORY
Exposed Server PC	The server PC is physically exposed	though separated from client PCs. Unauthorized access or tampering could disrupt operations.	High	Critical
The shop does not use a UPS system.	Doesn't utilize ups.	Sudden power outages can cause abrupt shutdowns, risking system corruption, data loss, and interruption of customer sessions.	High	High
Locked Assets	The internet cafe secures their peripherals	Physical assets (mouse, keyboard, PC components) are locked, reducing theft risk.	Low	Low
Hardware Malfunctions (Black Screen Issue)	The company experienced hardware issues.	The most damaging issue has been black screening caused by hardware malfunctions, leading to downtime and repair costs.	High	High



C. Operational & Policy Vulnerabilities

These vulnerabilities highlight risks associated with human behavior, improper system usage, and the absence of proper procedure. These issues significantly affect the organization's ability to maintain security, accountability, and effective incident response.

Table 4. Risk Assessment and Vulnerability Classification for Operational & Policy

FINDINGS		POTENTIAL IMPACT	VULNERABILITY	RISK CATEGORY
Inadequate Server Specifications	Regular PC used as the main server instead of enterprise hardware specifications.	Using a regular PC as the main server, which creates performance bottlenecks, reliability issues and security gaps.	High	Critical
No Redundancy in Network Infrastructure		A single router creates a critical single point of failure. If it fails, all 70 endpoints will lose network connectivity and internet access, resulting in a complete disruption of all operations.	High	High
Understaffed IT Team		Only 4 staff managing 70 workstations, server, network infrastructure, and data assets. Insufficient manpower for security monitoring, patch management, incident response, and user support limitations can create delays that increase the impact of security breaches and	High	High



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	prolong downtime.	system	
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VIII. BENCHMARKING ANALYSIS

The assessment of HG Esports iCafe highlights a business model centered on high-performance gaming equipment and customer experience. While the café demonstrates operational efficiency in delivering gaming services, its overall security and IT governance maturity reflects a reliance on informal practices rather than structured and standardized frameworks. This creates limitations in scalability, consistency, and long-term resilience.

A. Business Strategy

HG Esports iCafe demonstrates the presence of business and operational practices; however, documentation remains informal and inconsistently applied. Policies exist in practice but are not fully formalized or standardized, leading to gaps in communication, enforcement, and continuity. The absence of structured documentation reduces the organization's ability to maintain consistency and adapt to future operational demands.

B. Long-Term IT Strategy

The café shows evidence of long-term IT planning, particularly in maintaining updated systems and operational continuity. However, these processes are largely informal and undocumented. While planning is consistently practiced, the lack of formal structure limits adaptability to emerging technologies and reduces the effectiveness of long-term strategic alignment.

C. Security Policies

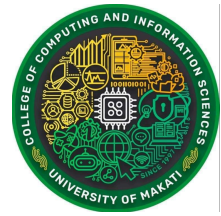
Security measures are implemented within HG Esports iCafe, but they are not supported by formal documentation or comprehensive enforcement. Critical controls such as firewall



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implementation and cloud compliance are either absent or insufficient. This creates exposure to external threats and reduces the overall effectiveness of the organization's security posture.

D. Information System Security Policy

The café enforces several security practices, including maintaining updated operating systems, utilizing reboot-to-restore software, and assigning individual employee accounts with defined privileges. Despite these strengths, significant gaps remain in multi-factor authentication, USB device control, and secure cloud storage practices. These limitations increase the risk of unauthorized access and data compromise.

E. Physical Access Controls

Basic physical security measures are present, including hardware protection and general environmental controls. However, the organization lacks advanced physical access mechanisms such as restricted access zones and comprehensive monitoring systems. This limits the ability to fully secure critical assets and infrastructure.

F. Network Management

HG Esports iCafe demonstrates partial implementation of network security controls, including system updates and basic maintenance. However, it lacks advanced protections such as properly configured firewalls, intrusion detection systems, and secure remote access mechanisms. These gaps increase vulnerability to cyber threats and unauthorized network access.

G. Network Security Controls

HG Esports iCafe demonstrates partial implementation of network security controls, including system updates and basic maintenance. However, it lacks advanced protections such as properly configured firewalls, intrusion detection systems, and secure remote access mechanisms. These gaps increase vulnerability to cyber threats and unauthorized network access.



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H. User Controls and Identity Management

User account management practices are present, particularly in password policies and employee offboarding. However, the absence of multi-factor authentication and automated account protection mechanisms weakens overall identity security. Strengthening authentication layers is necessary to reduce the risk of credential compromise.

I. Incident Management

The organization demonstrates the ability to respond to incidents and document issues as they arise. However, incident management remains reactive rather than proactive. Common issues, such as hardware malfunctions, are addressed only after occurrence, indicating a lack of preventive monitoring, testing, and formal escalation procedures.

J. Risk Management

HG Esports iCafe recognizes the importance of risk management and conducts periodic assessments and vulnerability scans. However, execution is inconsistent and lacks formal methodology. The absence of penetration testing, structured evaluation frameworks, and integration into strategic planning leaves the organization exposed to unmitigated risks.

K. Asset Management

The café operates a physical infrastructure consisting of approximately seventy gaming stations supported by a central server. Asset management is primarily focused on maintaining performance and usability for customers. However, the lack of formal asset tracking systems, logical segmentation, and scalable infrastructure limits operational efficiency and long-term growth.

IX. Conclusion



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The vulnerability assessment of HG Esports Cafe highlights a developing IT environment with several critical and high-risk security gaps, primarily driven by the absence of foundational network protections, reliance on informal policies, and limitations in infrastructure resilience. While the organization demonstrates strengths in maintaining updated systems, implementing basic user controls, and ensuring consistent service delivery through multiple ISPs, these measures are not sufficient to fully mitigate evolving cybersecurity threats. Key issues such as the lack of firewall protection, unrestricted USB access, exposed server infrastructure, and absence of a UPS system significantly increase the risk of malware infections, operational downtime, and potential data compromise.

HG Esports Cafe shows a clear gap in terms of formalized security frameworks, advanced technical controls, and comprehensive risk management practices. The café operates effectively from a performance and customer service perspective but lacks the structured governance and layered security approach observed in more mature organizations. These findings indicate that while HG Esports Cafe is functional and growing, its current security posture remains at a developing stage and requires immediate attention.

Overall, the results of this assessment provide a critical baseline for identifying vulnerabilities and prioritizing improvements. The insights gathered will support the development of a targeted Risk Control and Action Plan, enabling HG Esports Cafe to strengthen its technical defenses, formalize security policies, and enhance long-term operational resilience against both physical and cyber threats.



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APPENDICES

Exhibit A: Formal Interview Transcript

Table 5. Interview Transcript with HG Esports Cafe

Interview Questions	Client Response
Is the Windows OS and antivirus software up to date?	yes.
Does your shop use a firewall?	No.
What antivirus software are you using?	Not disclosed.
Is the main router still using the default password? (For example, was the Wi-Fi password never changed after installation?)	We already changed it.
Are users allowed to download different types of files?	yes.
Do users have any way or admin privileges that would allow them to download malware or bypass the timer?	No, only the admin can bypass it.
Over the years of operation, what has been the most frequent or most damaging issue experienced by the computer shop? (e.g., internet outages, hardware vandalism, virus infections, etc.)	I would say that the worst issue we have had so far is black screening, which was caused by hardware malfunctions.
Do all employees have access to the main server? Do they share one login, or do they have individual accounts with different privilege levels?	Not all of them, and they do not share accounts. Each account has its own privileges; for example, some accounts have hardware diagnostics access while others do not.



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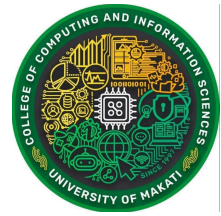
Do you have policies on what users are allowed or not allowed to download, tampering with PC settings, or accessing illicit content? If yes, how are these enforced by the staff?	Not disclosed.
Is the server PC used by employees for other activities such as web browsing, watching YouTube, or downloading files?	No, it is only used for the operations of the shop.
What software is used for the PC timer?	Our own software
When was the last time the OS and software of the PCs were updated?	Last week
Do you offer Wi-Fi access to customers? If yes, is it separated from the wired network used by the server and PCs?	Sometimes, if it is really needed, for example, when they need 2FA for logging into their accounts so we provide them with temporary Wi-Fi access on their phones. We also tell them to forget the network after they are done. No, we do not allow them to connect to the same network used by the PCs; we provide a separate internet network.
Who manages the main router? (e.g., which ISP—Globe, PLDT, etc.)	We have four ISPs: two are PLDT, and the other two are Converge and Globe.
Do you use reboot-to-restore software to erase user data and open tabs after restarting the PC?	yes.
Are users allowed to plug in USB devices? If yes, how do you prevent USB-based malware from spreading?	Yes. (No further details provided)
How do you ensure that the previous user did not install hidden monitoring or screen-capture software that could steal the next user's passwords?	We first ensure that the website they are downloading from is safe. They need to ask us first before downloading such files.
Where is the server PC located? Is it secured or exposed alongside the client PCs?	It is exposed but it is separated.



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How do you secure physical assets (e.g., preventing theft of mouse/keyboard or unauthorized opening of PCs to steal components)?	Yes, it is locked.
Do the PCs have an Uninterruptible Power Supply (UPS)? What happens in case of a sudden power outage? How are billing and remaining user time handled?	no.

Exhibit B: Photo Documentation



Figure 1. Documentation Of The Survey Questionnaire Provision for HG Icafe Comebo

Exhibit C: Specific Organizational Asset

Table 6. Specific Organizational Asset for HG E-Sports Cafe

HG E-Sports Cafe Assets List



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Assets	Hardware / Description	Quantity
Server Infrastructure	Server PC system unit (similar specs to regular PCs)	1
Network Infrastructure	On-Premise Servers	Available
Hardware and Networking Components	Router	1
	Switch	1
	Desktop PCs (Elite) (Ryzen 5 5500, 16GB RAM, GTX 1060, 27" Curved 180Hz FHD)	18
	Desktop PCs (VIP) (Ryzen 5 5600, 16GB RAM, GTX 1660 Super, 27" Curved 170Hz 2K)	25
	Monitors (S-VIP) (Ryzen 5 5600, 16GB RAM, RTX 5060 OC, 27" 240Hz FHD)	27
	Monitors (Elite) (27" Curved 180Hz Full HD monitors)	18
	Monitors (VIP) (27" Curved 170Hz 2K monitors)	25
	Monitors (S-VIP) (27" 240Hz Full HD monitors)	27
	Internet / ISP	4
	Keyboard (Magnetic Rapid Trigger Keyboard)	70



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	Headset (7.1 Badwolf Headset)	70
	Mouse (Rapoo Gaming Mouse (Elite & VIP), Badwolf Gaming Mouse (S-VIP))	70
Software	Operating System (Windows 11 Home installed on all units)	70
Personnel	Personnel	4
Data Assets	User session data and business records	(Not Specified)